Total No. of Questions : 10 ] [ Total No. of Printed Pages : 4

# Paper Code : 21317

## F-417

## B.C.A. (Fourth Semester)

## Examination, 2025

## (New Course)

## Paper -BCA-(N)-402

## Introduction to DBMS and SQL

*Time : Three Hours ]* *[ Maximum Marks : 70*

**Note** : Attempt any five questions. All questions carry equal marks. Symbol used have their usual meanings.

1. (a) What is database management system (DBMS)? What are the tasks performed by users in DBMS? What are the advantages and disadvantages of DBMS?



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(b) What is functional dependency? Explain its role in database design. Describe the inference rules for functional dependencies.

2. (a) What is relational model? Explain with example.

(b) Prove that BCNF is stricter than 3NF

3. (a) Who are data administrators? What are the functions of database administrator?

(b) Write short note on SQL. Explain various characteristics of SQL. What are the advantages and disadvantages of SQL.

4. (a) Define normal forms. List the definitions of first, second and third normal forms. Explain BCNF with a suitable example

(b) Discuss the following terms with suitable examples of each.

(i) DDL Command

(ii) DML Command

(iii) DCL



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5. (a) Discuss the architecture of DBMS. What are the type of DBMS architecture?

(b) Describe the classification of database language. What type of language is SQL?

6. (a) What are data models? Briefly explain different types of data models.

(b) Explain the various types of security and integrity threats in a database system. How can these threats impact data reliability?

7. (a) Give an over view of ORACLE architecture and features. How does ORACLE support transaction integrity?

(b) Describe different defense mechanisms used to protect a database system from security threats. Give examples where applicable.



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8. (a) How are constraints and indexes specified in SQl? Discuss the purpose and benefits of using them.

(b) Explain the anomalies that can occur in unnormalized database. How does normalization help eliminate them?



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